

Frontend – Assignment 1

Q1. HTTP Request–Response Cycle

Explain the **HTTP Request–Response Cycle** with a suitable diagram. Describe the major steps involved when a client (browser) requests a resource from a web server and receives a response.

Q2. Introduction to Web Technologies

Briefly explain the purpose and key features of the following web technologies:

1. HTML
2. CSS
3. JavaScript
4. Bootstrap
5. jQuery
6. React.js

Q3. Client-Side vs. Server-Side Technologies

What are **client-side technologies** and **server-side technologies**? Explain the difference between them with suitable examples.

Q4. Block-Level Elements in HTML

What are **block-level elements** in HTML? Explain their characteristics and provide at least **three examples**.

Q5. Semantic HTML Elements

What are **semantic elements** in HTML? Explain their importance in web development and provide suitable examples.

Q6. Creating a Table Using HTML

Write HTML code to create the following table:

Sr. No.	Name	Designation	Department
1	John	Manager	Sales
2	Smith	Sr. Manager	Marketing
3	Jane	Manager	HR

The table should include appropriate **table headings, rows, and columns**.

Q7. Creating a Nested List Using HTML

Write HTML code to create the following list using an appropriate combination of **ordered/unordered lists**:

- River
 - Ganga
 - Yamuna
 - Brahmaputra
 - Narmada

Q8. Types of CSS

Explain the different types of CSS used for applying styles to an HTML document. Describe each type with a suitable example.

Q9. Practical Understanding

Differentiate between **HTML, CSS, and JavaScript** based on their role in developing a web page. Explain what each technology contributes to a website.

Q10. Short Answer

Answer the following:

- a. What is the role of Bootstrap in web development?
- b. What is jQuery and why was it widely used in web development?
- c. What is React.js and what problem does it solve in frontend development?